

Wei Guo

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Research Interests

International Trade, Industrial Organization, Macroeconomics, Political Economy, Semiconductor Industry, Innovation

Education

Purdue University, West Lafayette, IN

Ph.D. in Economics

2022 – 2027(expected)

Boston University, Boston, MA

Shandong University, Jinan, China

Dissertation Committee

Mario J. Crucini (Chair)

Purdue University

Ralph Siebert

Purdue University

Chong Xiang

Purdue University

Anson Soderbery

Purdue University

Research

Job Market Paper

The Asymmetric Incidence of Intermediate Input Controls: Evidence from Semiconductor Export Restrictions

This paper studies the incidence of U.S. semiconductor export controls through trade, production, and strategic technology-capacity channels. I document large and heterogeneous responses in monthly semiconductor trade data: U.S. advanced-chip exports to China fall sharply after the October 2022 controls, while Japan, Taiwan, and Korea respond differently as alternative suppliers. I use these moments to discipline a unified four-country dynamic general equilibrium model with Armington chip sourcing, Leontief semiconductor bottlenecks, symmetric AI compute-capital accumulation, frontier-supplier market power, and endogenous allied compliance under access risk. The model shows that aggregate consumption welfare losses can be modest while strategic production-capacity losses are much larger. China cushions household consumption partly by cutting compute investment, which lowers future AI compute capacity. The access-risk mechanism further shows how allied non-compliance can erode suppliers' access to U.S. technology, amplifying welfare losses and reshaping global AI-capacity accumulation.

Publications

“Green Innovation and Maturity Mismatch: Evidence from China.”, with Zhang, Liqin, Xiao Sun, and Ruiqi Sun. *Finance Research Letters*, 2025, 85(C): 108124.

This paper examines how corporate green innovation affects firms’ debt maturity structures using panel data on Chinese listed firms from 2007–2023. I show that green innovation significantly exacerbates debt maturity mismatch, driven by long investment horizons, elevated financial risk, and short-term performance constraints. The results are robust to instrumental variable approaches, propensity score matching, entropy balancing, and policy shocks from carbon trading pilots. The findings reveal unintended financial risks associated with green innovation and inform green finance policy design.

Working Papers

“Social Credit Level and Corporate Leverage Manipulation: Evidence from China”, with Ruiqi Sun, Tianyu Zhang

Using China’s Social Credit System pilot as a quasi-natural experiment, this paper studies the impact of social credit on corporate leverage manipulation. Employing a difference-in-differences design on Chinese listed firms from 2007–2022, I find that higher social credit significantly reduces leverage manipulation. Mechanism analyses show that the effect operates through improved information transparency, expanded credit access, and reduced overleveraging. The effects are strongest among highly financialized, capital-intensive firms and firms facing lower financing costs.

Work in Progress

Multinational Semiconductor Firms and the Geography of Innovation, with Yuting Wei, Ralph Siebert

This project studies the geographic allocation of R&D by multinational firms in the semiconductor memory industry. Using firm-level production, capacity, plant, and patent data for the DRAM industry from 1974–2004, we document that innovation is concentrated among multinational firms, that firms replicate internal technology structures across countries, and that R&D entry increases local knowledge spillovers. We develop a multi-country endogenous growth model in which heterogeneous firms choose both the scale and location of R&D. The model is used to assess whether firms under-expand geographically relative to the efficient allocation and how policies such as the U.S. CHIPS and Science Act affect industry growth, productivity, and the global stock of knowledge.

The Transition from Gas Cars to EVs, with Federico Rossi, Ting Zhu

This project studies how electric-vehicle adoption reshapes the retail gasoline market through consumer demand, station pricing, and exit decisions. We combine daily prices for roughly 1,700 Chicago-area gas stations with a geolocation panel of about 150,000 consumers, whose individual station visits we construct from raw GPS records, and estimate a random-coefficients demand model in which consumers trade off price against the detour a station requires from their daily travel. We use the model to simulate pricing and exit as EV penetration increases: heterogeneous adoption produces station closures and fuel deserts in some areas and higher prices through reduced spatial competition in others. The project evaluates counterfactual policies, such as charging-infrastructure expansion, and their distributional welfare implications.

The Dynamic Propagation of Critical-Input Wedges in Production Networks: Short-Run Bottlenecks, Long-Run Catch-Up, and the Conditions for Backfire

This project studies how restrictions on critical upstream inputs propagate through production networks and induce endogenous domestic capability responses. I develop a dynamic multi-sector model in which an input wedge generates short-run welfare losses through production bottlenecks but also induces domestic production and learning-by-doing in the targeted input sector. The model characterizes when long-run capability gains can outweigh short-run bottleneck losses and how this trade-off depends on input substitutability, network exposure, and the strength of learning. The framework is motivated by recent semiconductor export controls and other episodes of technology restrictions.

Teaching Experience

Instructor: ECON 251 Microeconomics <i>Distinguished Teaching Winner</i>	Sep 2025 – Dec 2025 Purdue University
Instructor: ECON 251 Microeconomics <i>Distinguished Teaching Winner</i>	Summer 2025 Purdue University
Teaching Assistant for Game Theory(ECON 510) <i>TA to Prof.Ralph Siebert</i>	Jan 2026 - May 2026 Purdue University
Teaching Assistant for Real Estate Fundamentals(MGMT 370) <i>TA to Prof.Lindsay Relihan</i>	Sep 2023 - May 2024 Purdue University
Teaching Assistant for Macroeconomics(ECON 352) <i>TA to Prof.Cathy Zhang</i>	Sep 2022 - May 2023 Purdue University

Conference Presentation

AEA 2027 Annual Meeting, Washington, DC,	Spring 2027 (scheduled)
SEA 96th Annual Meeting, Houston, TX	Fall 2026 (scheduled)
JIE Summerschool and JIE Editors conference, Milan, Italy,	Summer 2026
HEC Economics Ph.D Conference, Paris, France	Summer 2026
MEA's 90th Annual Meetings, Chicago, IL	March 2026
SEA 95th Annual Meeting, Tampa, FL	Fall 2025
14th Annual Research Design for Causal Inference, Northwestern University, IL	Summer 2025
Estimation of Structural Models with Conditional Choice Probabilities, University of Notre Dame, IN	Summer 2025

Programming and Data Skills

MATLAB, Stata, Python, R, SAS, SPSS, Excel (VBA), L^AT_EX
Large-scale administrative data, geolocation data, structural estimation, simulation-based methods

Honors and Awards

Economics Research Micro-Grant Fund (\$700), Purdue University	2025
Distinguished Teaching Winner	Summer of 2025
Distinguished Teaching Winner	Fall of 2025

Professional Service

Referee for peer-reviewed academic journals (Elsevier).

Additional Experience

Research Assistant, Purdue University 2022 – Present

Business Doctoral Student Association Treasurer Fall & Spring 2023

Industry experience in financial analysis and investment research (details available upon request)